

WEEK 4 PROJECT SUMMARY

HealthConnect Experience Lab — Data Analytics Track

Nancy Lee YIMBERE ALAPINI | Performance & Decision Intelligence Analyst

WEEK 4 TAKEAWAY

Week 4 establishes the analytical foundation of the HealthConnect project: understanding the appointment data, assessing its quality, defining the business questions to investigate, and preparing a structured approach to analyse no-shows without drawing premature causal conclusions.

1. SPECIFIC PROBLEM ADDRESSED BY THE DATA ANALYTICS TRACK

HealthConnect Clinic seeks to better understand why some scheduled appointments are missed and which observable patterns in the available data may help guide further investigation and future operational decisions. The Week 4 role of the Data Analytics Track is therefore to structure this problem from the available data, without yet building a dashboard or producing final analytical conclusions.

2. RESOURCES USED

- HealthConnect Appointment Dataset — the main file containing appointment-level records.
- HealthConnect Data Dictionary — the reference used to understand the meaning and analytical role of the variables.
- Week 4 Assignment — the official AnalystLab Africa project brief and submission requirements.
- Mentor Brief — clarification of the Week 4 scope, expectations, and Data Analytics deliverables.

3. KEY OBSERVATIONS FROM WEEK 4

The initial inspection confirms that the dataset is structured at the appointment-record level, with `appointment_id` serving as the record identifier. The initial data quality assessment also identified issues that must be controlled before analysis: limited missing values in selected operational variables, structural missingness associated with reminder information, and inconsistencies in some demographic or historical attributes linked to repeated `patient_id` values.

These observations do not prevent appointment-level analysis, but they limit the reliability of longitudinal patient-profile reconstruction. Historical and demographic variables will therefore require cautious interpretation.

4. PROPOSED APPROACH

The future analysis will follow a progressive sequence: first establish the overall extent of no-shows, then investigate how no-show patterns vary across appointment and scheduling context, reminder practices, accessibility and operational context, recorded patient history, and recorded patient characteristics.

The analysis will be guided by the Business Questions and analytical hypotheses defined during Week 4. Planned methods include distributions, proportions, rate comparisons, cross-tabulations and, only where justified, appropriate statistical tests. Observed relationships will be treated as associations unless the available analytical design supports stronger causal inference.

5. KEY CONSIDERATIONS AFFECTING THE PROJECT

CONSIDERATION	ANALYTICAL IMPLICATION
Analytical grain	Maintain the analysis at the appointment-record level.
No-Show Rate	Finalize the eligible population and denominator before calculation.
Patient history	Do not reconstruct unreliable longitudinal history from patient_id.
Reminder analysis	Compare reminder channels only among appointments where a reminder was sent.
Continuous variables	Create analytical groups only where analytically or operationally justified.
Interpretation	Association does not imply causation; future recommendations must remain evidence-linked.

6. PROPOSED FOCUS FOR WEEK 5

Subject to the official Week 5 instructions, the logical next step will be to prepare the data for analysis, finalize metric definitions and analytical populations, and begin answering the Business Questions and testing the proposed hypotheses. The three potential KPIs identified — No-Show Rate, Reminder Coverage Rate, and No-Show Rate by Reminder Status — will be evaluated within this framework, while additional segmentation measures will remain analytical metrics unless a clear decision or monitoring need justifies their promotion to KPI status.

CONCLUSION

Week 4 is not intended to provide a definitive explanation of no-shows. Its purpose is to establish a reliable analytical foundation for the next stages of the HealthConnect Experience Lab. The work completed provides a structured, traceable framework for moving from the raw appointment dataset to evidence-based investigation while maintaining explicit boundaries around data quality and interpretation.

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